

SOURCE TO SOUND

DRAIN DETECTIVES

Map the storm drains in your school to discover where your school's pollution travels and how it reaches local waterways.

WHAT ARE STORM DRAINS?

A storm drain is the metal grate you see at the edges of roads and parking lots.

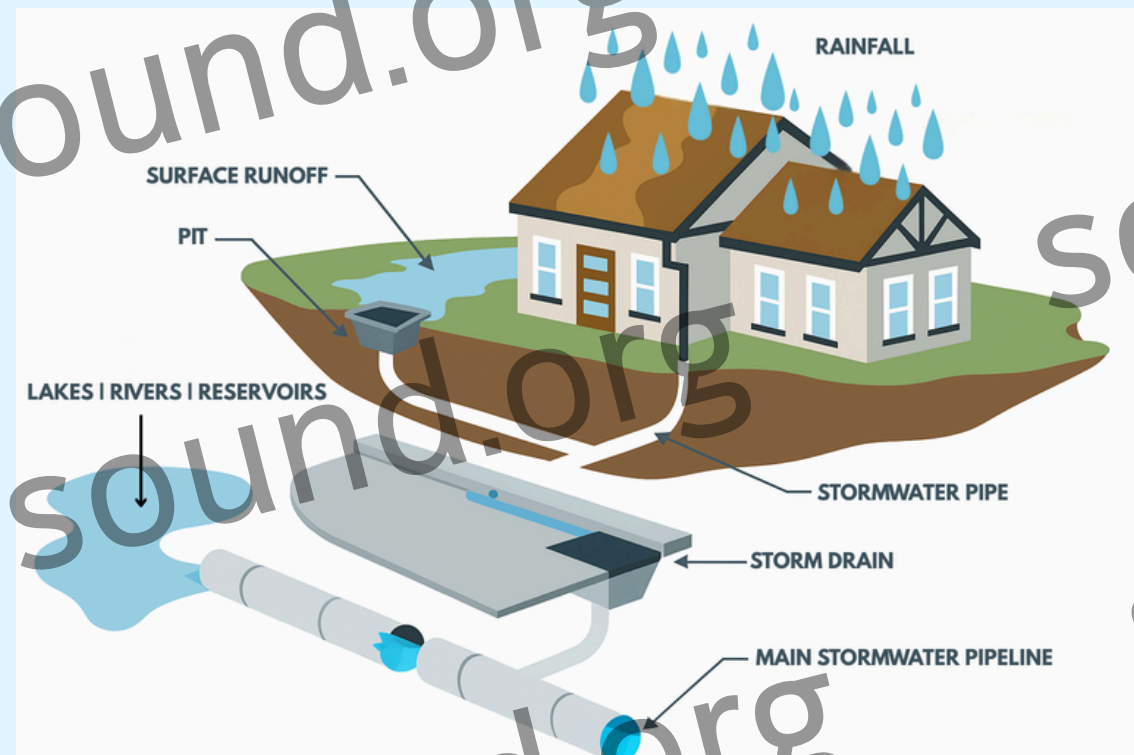
These are underground drainage systems designed to collect and redirect surface runoff, the flow of excess water across roads and other paved surfaces.

Unlike sinks and household drains, storm drains bypass water treatment facilities. Instead, water flows directly into lakes, rivers, and streams without treatment or filtration.



FROM STORMS TO STREAMS

Every storm drain in your school connects to a stream



When it rains, pollution from your school flows into storm drains, and straight into the stream. This includes:

- Oil from car engines in parking lots
- Fertilizer containing nitrogen, phosphorus, and pesticides from fields
- Food waste runoff and cleaning chemicals from cafeteria areas
- Paint chemicals from painted surfaces and courts
- Litter, dirt, and waste from general foot traffic

Organisms living in lakes and streams, such as Coho and Kokanee salmon, are highly sensitive to these pollutants. They can disrupt their reproduction and deplete oxygen levels.

LOW IMPACT DEVELOPMENT

Low Impact Development, or LID, are design strategies that manage polluted stormwater at the source, by slowing, filtering, and allowing it to soak into the ground instead of rushing into a drain.

RAIN GARDEN

A shallow planted depression that collects runoff from rooftops or roads and lets it infiltrate the soil.



BIOSWALE

A planted channel that slows, filters, and redirects stormwater as it moves across a surface.



PERMEABLE PAVEMENT

Paving material that allows water to pass through it directly into the ground below.



NATIVE PLANT STRIPS

Vegetation placed between paved surfaces and drains that absorbs and filters runoff.



YOUR TASK

For every storm drain you find, document the following:

Drain Number: Start at 1 and go up

Location: Street name or intersection

Connected Waterway: Which body of water does the drain connect to?

Condition: Is there visible oil, sediment, debris, or discoloration at the drain inlet? Describe it!

Low Impact Development: Is there an LID feature between the road and the drain? Describe it!

Estimate distance to nearest body of water

Photograph of the drain

As a Drain Detective, your task is to locate the drains around your school, understand which bodies of water they connect to, and figure out whether there are any filtration systems in place to stop stormwater pollution before it reaches them.

CONNECTING DRAINS TO STREAMS

Here are some tools to help you find which body of water a stormwater drain connects to:

OpenStreetMap: Free, global, browser-based map where students can search their school address, zoom in, and see roads, streams, and waterways labeled.

Google Maps: Search a school, switch to satellite or street view, and trace the path from a drain to the nearest visible water body.

EPA's How's My Waterway: US-only website where students type in their location, and see nearby waterways with water quality data attached.

Ask Around! Consider asking school facilities managers, city stormwater departments, teachers, or neighbors what they know.

BUILDING A CLASS MAP

Back in the classroom, each team plots their drains on a large printed map of the school. Each drain gets a numbered pin.

Color code them:

Green Pin: Low Infrastructure Development present

Red Pin: No filtration methods present

Draw lines from the pins to their connected streams.

Once the map is complete, answer these questions, and begin a class discussion:

How many drains did you find total?

How much are unfiltered? How many have LIDs?

Which streets/areas have the highest concentration of unfiltered drains?

Which drain sits closest to a salmon bearing stream with no filtration?

If the city could immediately fix 1 drain, which would you suggest, why, and how? Describe new policies or ideas you would implement.